

**5DATA004C Data Science Project Lifecycle
(2025/26)**

Group Coursework

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Table of Contents

Project Plan	3
1.1 Requirements and Expected deliverables	3
1.1.1 Expected Deliverables	3
1.2 Project Timeline	4
1.4 Identification of Potential Risks and Mitigation Plan.....	5
2. Background and Problem Statement	7
2.1 Background	7
2.2 Problem Statement	7
2.3 Objectives	8
3.Data and Methods	8
3.1 Data Presentation	8
3.2 Data Quality issues identified	10
3.3 Methodological Approach	11
3.4 Tools and Environment	12
4. Analysis and Recommendations	13
4.1 Exploratory Data Analysis (EDA)	13
4.2 Statistical Correlation and Multicollinearity	15
4.3 Predictive Modelling	16
4.4 Ten-Year Projection.....	17
4.5 Actuarial Translation and Recommendations.....	18
5. Individual Reflection	20
5.1 What I Learned	20
5.2 Challenges Faced and How I Overcame Them	20
6.Evidence of Work	21

Project Plan

1.1 Requirements and Expected deliverables

This project is being commissioned by Aethelgard Re, an imaginary global reinsurance company, because they currently use historical averages for their underwriting model, and are ignoring the rate of change of regional temperature anomalies. Reinsurers take losses from primary insurers which exceed their limits, and ultimately they rely on the assumption of the climate in the risk models they use to determine their pricing. My brief is for one technical report in relation to four functional requirements as the only consultant on this deferred/referred submission.

Exploratory Data Analysis (EDA): Clean data from six climate/emissions datasets, deal with anomalies, and visualize historical trends in temperatures and warming drivers.

Statistical correlation: detect possible collinear warming factors and to directly correlate their relationship with temperature increases.

Predictive modelling: train a baseline regression model for temperature anomalies over the next decade, split the data appropriately and assess it with traditional model evaluation metrics (MAE, RMSE, R^2).

Actuarial translation: translate statistical results and present the conclusions and recommendations for adjusting premiums and allocating assets for a non-technical audience.

Three non-functional constraints in the brief, that influenced the scope of the work, were as follows:

- it had to be done as a single student project, not a team project;
- it had to be delivered within a compressed deferred/referred scope (rather than the scope of the original semester); and
- it had to be usable by two types of audience at once; the statistically minded module team marking it, and implicitly, the non-statistically minded Aethelgard Re executive reading the actuarial recommendations.

1.1.1 Expected Deliverables

- One Word/PDF report that includes: project plan, background/ problem statement, data/methods, analysis/recommendations, individual reflection, and evidence of work.

- A publicly available GitHub repository with the entire Python analysis pipeline, from the six raw data files and the data loading, cleaning, exploratory data analysis, correlation/VIF diagnostics, regression modelling and projection.
- At least six data visualisations (trend charts, a correlation heatmap, and model-fit/projection charts) embedded in the report with captions and in text discussion, with each figure carrying a certain claim, and not being used just for decoration.

1.2 Project Timeline

Given the compressed deferred/referred timeline (handed out 16th June 2026, due 10th July 2026), the work was planned and executed across four working days, with each day producing a concrete, checkable output rather than open ended activity:

Day	Key Tasks	Deliverable Produced	Estimated Duration
Day 1	▪ Load, clean and merge all six datasets to World/Global level ▪ resolve data quality issues ▪ initial EDA and trend plots	merged_global.csv; 3 trend charts	3-4 hours
Day 2	▪ Correlation matrix ▪ VIF multicollinearity diagnostics ▪ Feature selection rationale	Correlation heatmap and VIF table	2 – 3 hours
Day 3	▪ GitHub upload ▪ Baseline regression modelling (4 candidate specifications) ▪ train/test evaluation ▪ 10-year projection ▪ actuarial translation	▪ Make public GitHub repo and upload data files and code notebook (py) ▪ Model comparison table ▪ Projection Chart ▪ Recommendations	3-4 hours

Day 4	Report writing	Writing about plan, background	2-3 hours
Day 5	Report writing	Writing about data methods, analysis, recommendations	3-4 hours
Day 6	Report writing	Individual reflection	2-3 hours
Day 7	Finalizing the work	Checking final report, GitHub upload and code annotation, proofreading	4-5 hours

1.4 Identification of Potential Risks and Mitigation Plan

Risks were assessed on simple likelihood/impact basis, consistent with standard project risk registers, and mitigations were built into the plan above rather than treated as contingencies to action later

Risk	Likelihood	Impact	Mitigation
Multicollinearity among warming drivers distorts regression coefficients	High	High	Compute a correlation matrix and VIF (Section 4.2) before modelling; select a single, non-derived driver (GHG emissions) rather than combining collinear attribution variables.
Compressed deadline for an individually adapted, originally group-scoped brief	High	Medium	Scope the technical work to the World/Global aggregate rather than all 190+ countries, keeping the pipeline fast to build, test and validate within four days.
Baseline model may fit poorly given accelerating, non-linear warming	Medium	Medium	Treat weak test set performance as a genuine, reportable finding rather than a failure to be hidden; investigate residuals rather than only the summary metric (Section 4.3).

Data quality issues (missing codes for aggregate rows, inconsistent year ranges/extra columns across files)	Medium	Low	Filter to the 'World'/'Global' entity rows consistently and explicitly select value columns by name (not by position) across all files; verify merged data has zero missing values before modelling.
Report becomes too technical for the non-technical audience the brief specifies	Medium	Medium	Separate statistical detail (Section 4.1-4.4) from a dedicated, plain-language actuarial translation subsection (Section 4.5) aimed explicitly at non-technical readers.
Working solo removes the peer cross-checking a group project would normally have	Medium	Low	Cross-check every regression result against multiple candidate model specifications (Section 4.3) rather than accepting a single model's output at face value.

2. Background and Problem Statement

2.1 Background

Aethelgard Re is a global reinsurer and plays a significant part in helping primary property insurance companies to stabilise the insurance market by absorbing high-value losses associated with catastrophic events. The firm has seen a dramatic increase in claims in recent years in connection with multi-billion dollar events related to extreme weather and extended regional temperature swings. It reflects a real-world trend in the industry, as Swiss Re Institute's sigma research projects that natural cat values in the world will be at around USD 137 billion, with economic losses at USD 318 billion, creating a vast uninsured protection gap; the same research also reveals that insured natural cat values have increased at a steady rate of 5-7% annually in real terms over the last few years, as a result of a combination of asset increase, urbanization and increasing intensity of natural catastrophes.

Swiss Re's analysis also finds that a significant but not yet dominant piece of this loss growth is due not just to exposure growth, but to intensification of hazards linked to climate change, and separately estimates that insured losses from natural catastrophes may double globally in about 10 years if trends continue. The Intergovernmental Panel on Climate Change (IPCC) has also reached the same conclusion in its Sixth Assessment Report cycle to date, that human greenhouse gas emissions have clearly caused global warming, and that more warming will lead to more extreme weather events. These sources corroborate the client's main argument: that relying on historical averages is a less reliable method for precisely pricing catastrophic risk, as the underlying climate system is not stationary.

Underwriting catastrophic risk via historical averages is reasonable, provided the underlying climate system is stable over time, but is not an appropriate choice in a changing climate. The current underwriting approach of Aethelgard Re is to write the catastrophic risk on historical averages, which is a reasonable approach only if the underlying climate system is stable over time. The evidence presented in this report indicates it is not: global warming is not on a long term trajectory, it seems, but is instead accelerating (Section 4.3), with implications on the ability of a reinsurer to rely on historical baselines for future risk pricing.

2.2 Problem Statement

The big question is, which particular atmospheric warming drivers are the closest to rising temperature anomalies (e.g. CO₂, methane, nitrous oxide, fossil fuel emissions, land use emissions) and whether they can be reasonably used to produce a predictive model for warmer temperatures, and what this bodes for premium pricing and asset allocation over a 10 year timeframe? This is very much in keeping with the project brief, EDA, correlation/multicollinearity analysis, baseline regression modelling with appropriate train/test

validation, and an actuarial translation of results. The problem is deliberately bounded to World/Global-level analysis (rather than country-by-country risk zoning), since this is a personal, time-specific submission and regional/country level analysis is a matter of scope but is noted as a natural extension in Section 4.5.

2.3 Objectives

The problem statement is broken down into specific, measurable, attainable, relevant and timely (SMART) outcomes in this report:

- Objective 1: Clean and visualise six historical data sets of climate and emissions to be able to see long time series of warming drivers (completed in section 4.1).
- Objective 2: Calculate the amount of correlation between each driver and the global temperature anomaly; formally check for multicollinearity using a correlation matrix and the Variance Inflation Factor (VIF) analysis (which is delivered in Section 4.2).
- Objective 3: Using an earlier period of data to train a base model and a later period of unseen data to validate the model, using MAE, RMSE and R^2 to assess the model and comparing to at least three alternative models (delivered in Section 4.3).
- Objective 4: Forecast the temperature anomaly over a decade, and present the statistical results as concrete evidence-based premium and asset-allocation advice for a non-technical user (in Sections 4.4-4.5).

3.Data and Methods

3.1 Data Presentation

Dataset	Content	Granularity
temperature-anomaly.csv	Global average temperature anomaly vs. 1961-1990 baseline, with 95% CI bounds	Global/hemisphere, 1850-2021
contribution-temp-rise-degrees.csv	Total temperature-rise contribution attributable to GHG emissions	Country + World, 1851-2021
contribution-to-temp-rise-by-gas.csv	Temperature-rise contribution split by CO ₂ , methane and nitrous oxide	Country + World, 1851-2021
global-warming-fossil.xlsx	Temperature-rise contribution from fossil fuels and industry	Country + World, 1851-2021
global-warming-land.xlsx	Temperature-rise contribution from agriculture and land use	Country + World, 1851-2021

Dataset	Content	Granularity
ghg-emissions-by-world-region.xlsx	Annual GHG emissions in CO ₂ -equivalent tones	Country + World, 1850-2021

After filtering to the World/Global aggregate row and merging on Year, the resulting analysis ready panel covers 171 annual observations (1851-2021) across 8 numeric variables, with zero missing values. Table below summarizes the distribution of each variable:

Variable	Mean	Std. Dev.	Min	Max
Temperature anomaly (°C)	-0.082	0.365	-0.598	0.933
N ₂ O contribution (°C)	0.022	0.024	0.0001	0.083
Methane contribution (°C)	0.130	0.127	0.0004	0.415
CO ₂ contribution (°C)	0.318	0.304	0.001	1.113
Fossil-fuel contribution (°C)	0.245	0.292	0.0003	1.062
Land-use contribution (°C)	0.224	0.168	0.001	0.548
GHG emissions (Gt CO ₂ -eq)	19.8	15.6	4.1	54.8
Total GHG contribution (°C)	0.469	0.454	0.002	1.611

There are two things that stand out from this summary. First, the standard deviations are large relative to the means for almost every driver, consistent with the strong long run upward trend visible in Figures 1-3 (early period values near zero, late period values well above the mean). Second, GHG emissions and the derived contribution variables span very different scales (billions of tones vs. fractions of a degree), which is why all regression features were used on their native scale with an intercept term rather than requiring manual rescaling for the linear models used here.

3.2 Data Quality issues identified

The most significant data quality problems identified were Rows for continents and aggregates (e.g. 'World') have no ISO 'Code', so a naive filter on non-null codes would drop the rows required for a global level analysis without notifying. The pipeline on the other hand filters on the 'Entity' column ('World'/'Global').

The files are temperature_anomaly.csv, which has three value columns (the anomaly and 95% confidence interval upper and lower bounds), and the other five, each with a single value column. The original early version of the data loading function had assumed a uniform structure for all six files and failed with a length mismatch exception; the solution was to specify the column names explicitly of the files to be loaded, rather than attempting to guess them positionally, which resulted in another concrete example of a data-quality assumption that had to be explicitly addressed (see also Section 5.2).

Since each file has a slightly different year range and number of rows at country level (roughly 37,500 to 41,700 rows), an inner join on Year was performed to avoid creating missing values in the merge operation, which would have been done using a left/outer join. Note that the values of contributions are very small (e.g. 10^{-6} °C for value in early 19th) as expected with very low industrial era emissions at this time, but there is a need to be careful with floating-point rounding when printing output tables.

‘World’ is already pre aggregated, and it is even published directly in each of the source files, not a rollup of the sum of all countries; this means that there is no need for a double counting error as a result of overlapping regions (e.g., if one country is included in a continental rollup and another is included in a global rollup).

3.3 Methodological Approach

Each of the six datasets was selected to include only the 'World'/'Global' rows before being joined together using Year as the key. The plan was then to have four stages, as depicted in Figure 1 below:

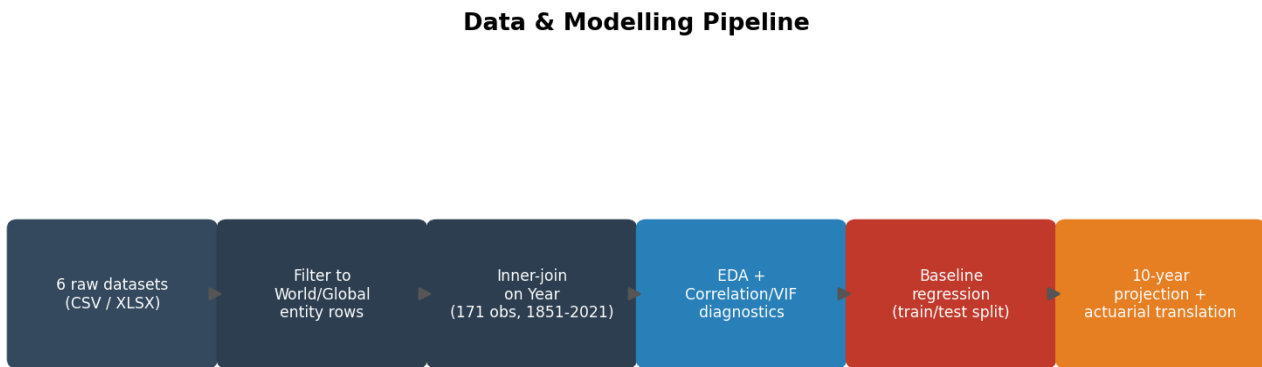


Figure 1: End-to-end data and modelling pipeline used in this project

EDA: visualisation of long-run trends and inflection points with time series plots of the temperature anomaly, gas specific contributions and total GHG emissions (Section 4.1).

Multicollinearity diagnosis: a Pearson correlation matrix across all candidate drivers and the target, and Variance Inflation Factors (VIF) computed using statsmodels, are used to test the brief's expectations of a high degree of interrelatedness among warming drivers (Section 4.2).

Baseline regression: a temporal train/test split regression model for this problem (training on 1950-2005 and testing on 2006-2021) implemented by scikit-learn's LinearRegression with Mean Absolute Error (MAE), Root Mean Squared Error (RMSE) and R^2 . A series of candidate specifications were evaluated and compared and a final specification was selected (Section 4.3).

Projection: the selected model together with the recent trend (2000-2021) in the chosen driver, were used to project the temperature anomaly 10 years into the future (to 2031) (Section 4.4). The suggested split (train 1980-1990, test 1991 onward) was determined by the number of annual observations, which was just 171; an 11 year training window was deemed too short to estimate a stable trend, especially since warming has accelerated significantly since the mid-20th century. The window was thus extended to 1950-2005 (56 training years) with a test window of 2006-2021 (16 years), which is a preservation of the intent of the brief (a short-term training period followed by a long period of an unseen climate) sufficient to provide the model with enough data to fit.

3.4 Tools and Environment

The analysis was implemented entirely in Python, run via Google Colab against source files stored on Google Drive so that the environment requires no local setup.

Tool / Library	Role in the Pipeline
pandas	Loading CSV/XLSX files, filtering, merging, and tabulating descriptive statistics
matplotlib	All EDA trend charts, the correlation heatmap, and the model-fit/projection charts
scikit-learn	Linear Regression modelling, Polynomial Features, and MAE/RMSE/R ² evaluation
statsmodels	Variance Inflation Factor (VIF) computation for multicollinearity diagnosis
Google Colab + Google Drive	Execution environment and shared file storage for reproducibility

4. Analysis and Recommendations

4.1 Exploratory Data Analysis (EDA)

The global temperature anomaly (relative to the 1961-1990 baseline) shows a clear long run rise, from roughly -0.23°C in 1851 to $+0.76^{\circ}\text{C}$ in 2021, with the steepest, most consistent increase occurring after the 1970s-1980s.

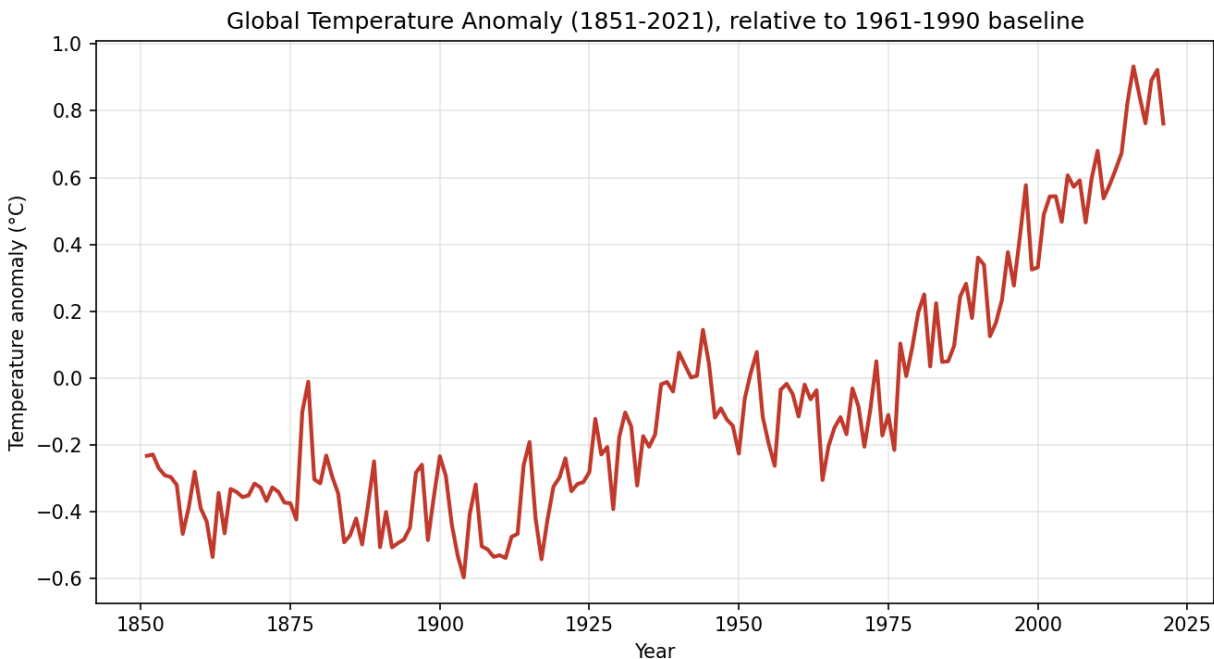


Figure 2: Global temperature anomaly, 1851-2021(relative to 1961-1990 baseline)

Annual Global GHG emissions increased from about 4.1 billion tonnes CO_2e in 1851 to more than 54 billion tonnes CO_2e in 2021, and increased at the fastest rate since the mid 1900s. Separating the contribution by gas type (not shown here in full) reveals that CO_2 is overwhelmingly the major contributor, methane is a secondary contributor, and nitrous oxide is the smallest of the three contributors with all three rising and falling nearly in perfect synchrony a strong visual cue that this is a case of multicollinearity, which was formally tested in Section 4.2.

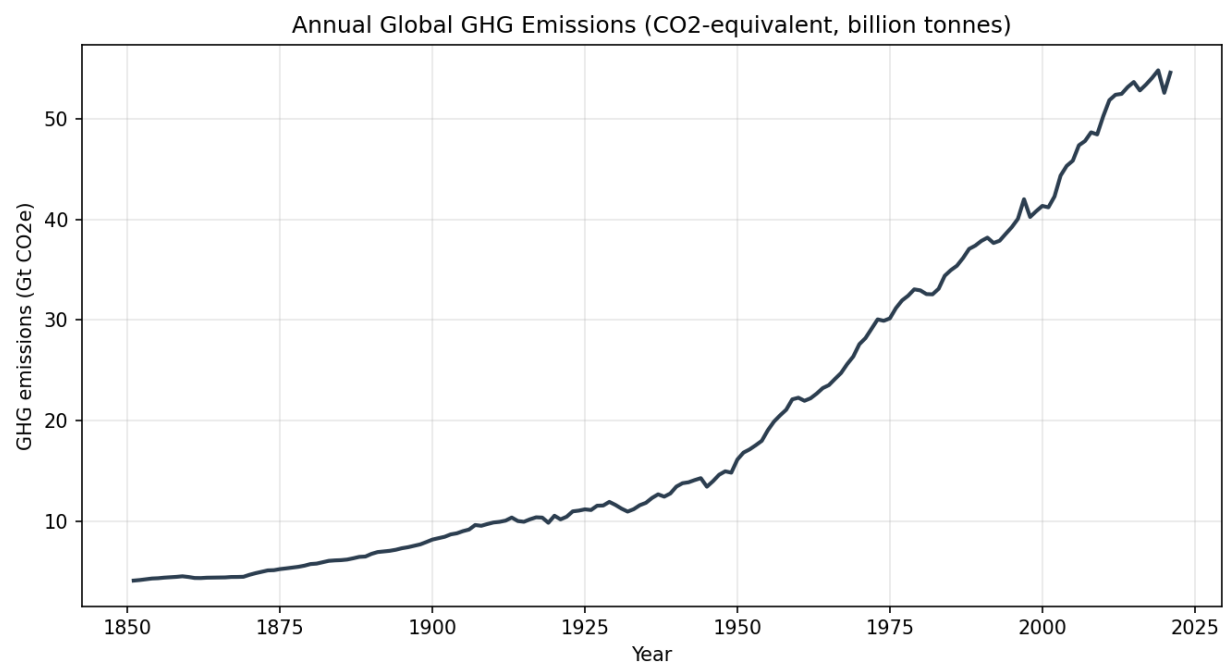


Figure 3: Annual global GHG emissions (CO₂ equivalent), 1851-2021

Put differently, emissions of all types have increased over the past more than 100 years, and so have temperatures, at a higher rate since about the 1970s. The reason a simple historical average pricing model is unable to keep up (Section 4.3) is the single observation, that the recent past looks nothing like the 19th or early 20th century.

4.2 Statistical Correlation and Multicollinearity

All warming drivers have strong correlations with the temperature anomaly: from 0.89 to 0.94 (Pearson) for all of them. Critically, the drivers are also nearly perfectly correlated with each other (pairwise r almost always greater than 0.94, and more than 0.99 for several pairs, like CO₂ and methane contributions). This collinearity is serious several orders of magnitude greater than the common $VIF > 10$ threshold for serious collinearity is confirmed by a formal Variance Inflation Factor (VIF) analysis.

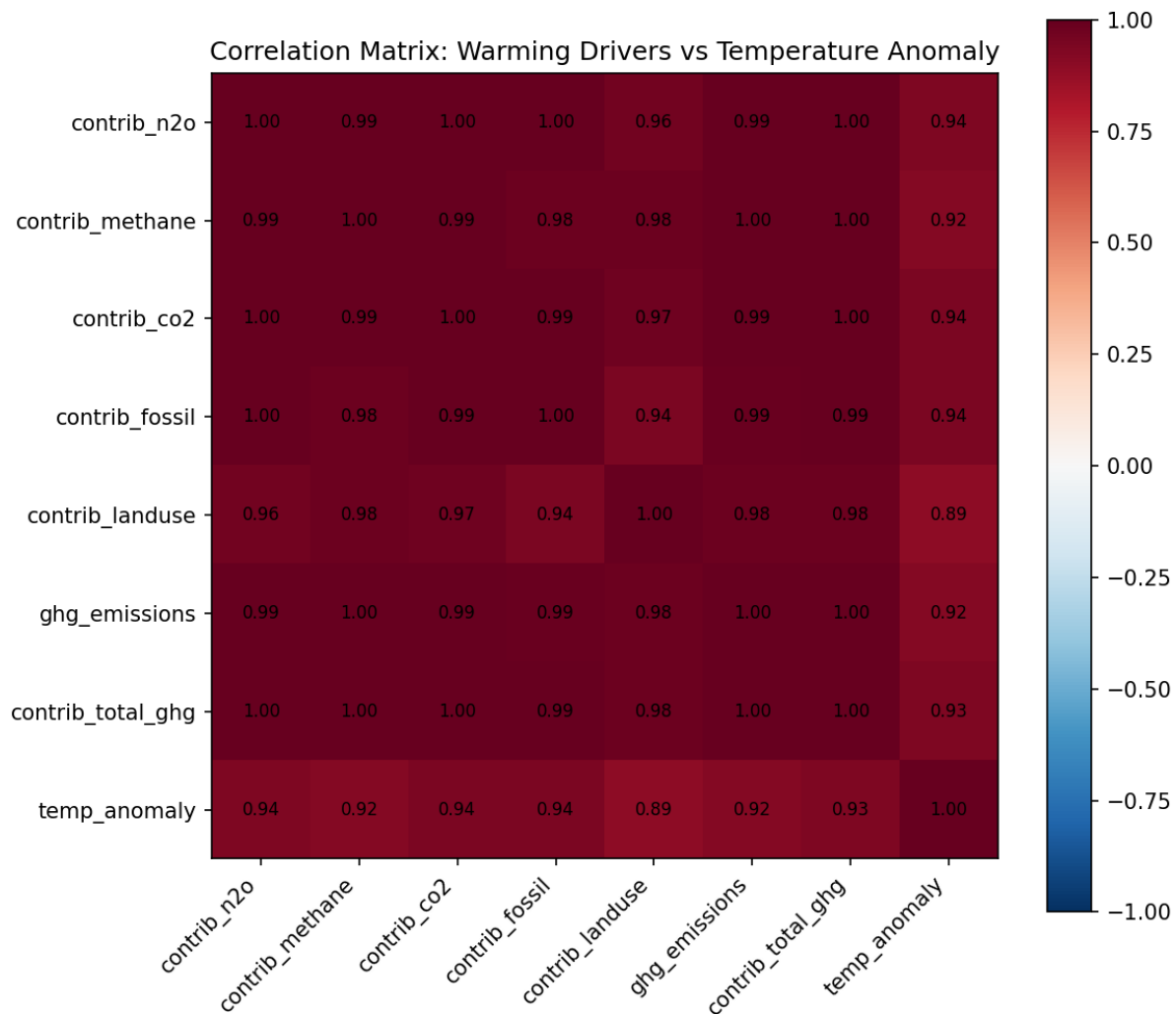


Figure 4: Correlation matrix of warming drivers and temperature anomaly

Driver	Correlation with temp. anomaly	VIF
Fossil-fuel contribution	0.939	4.5×10^9
CO ₂ contribution	0.938	4.9×10^{10}
Nitrous oxide contribution	0.936	1.4×10^7
Methane contribution	0.919	1.6×10^9
Land-use contribution	0.891	2.8×10^9
GHG emissions (CO ₂ -eq)	0.918	220

It is expected: the five contribution variables are all statistics calculated from the same underlying emissions series, and this would be circular and produce coefficients that are unstable and uninterpretable when all five are put in one regression just what the project brief warns of. Therefore, GHE (raw CO₂-equivalent tones) was kept as the only regression predictor, as this is the raw causal effect instead of a derived attribution, has a much lower (but still high) VIF, and is a lever that Aethelgard Re's counterparties can actually leverage emissions reduction.

4.3 Predictive Modelling

Four candidate baseline specifications were compared, all trained on 1950-2005 and tested on 2006-2021:

Model	MAE (°C)	RMSE (°C)	R ² (test)
Year (simple linear trend)	0.166	0.194	-0.840
GHG emissions (linear)	0.132	0.162	-0.287
GHG emissions (quadratic)	0.381	0.404	-7.000
Year + GHG emissions	0.225	0.245	-1.933

The GHG emissions linear model produced the lowest error (MAE 0.13°C, RMSE 0.16°C) and was retained as the final baseline model: temperature anomaly $\approx -0.642 + 2.37 \times 10^{-11} \times \text{GHG emissions}$. Every specification, however, produced a negative R² on the test set.

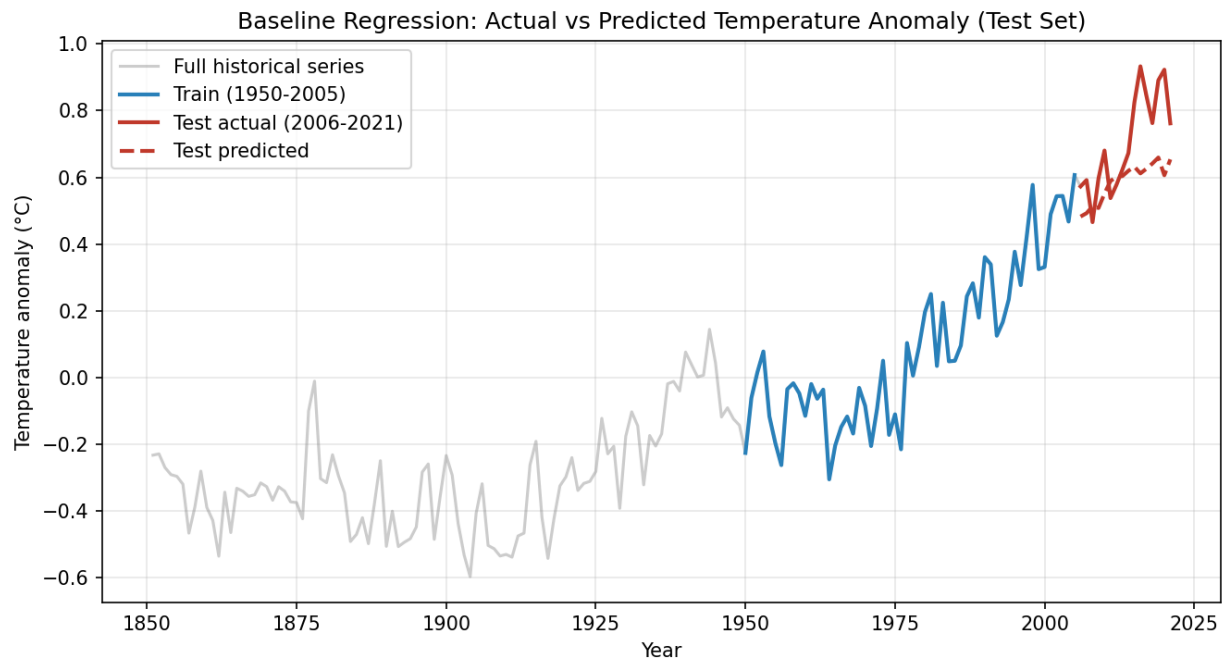


Figure 5: Baseline model: actual vs predicted temperature anomaly on the 2006-2021 test set

This is a true discovery, not a modelling mistake. If R^2 is negative, the model does not explain any of the variance in test period data the model is worse than predicting the average of the test period data. Why, because when the residuals are plotted, the model (which was fitted to the 1950-2005 relationship) consistently underestimates the 2006-2021 period. This means that even the warming rate with respect to emissions has increased since 2005, so that the relationship has become not linear but accelerated. The model has an R^2 below zero, but this systematic bias is small compared to the year-on-year variance in the test set due to natural variations such as El Niño/La Niña, which leads to a small reduction in variance in the test set but is more than outweighed by the systematic bias.

The most critical technical discovery in this report is the baseline linear model's systematic underestimation of recent and future warming, which buttered the brief's hypothesis that an average from the past no longer serves as a valid baseline to price.

4.4 Ten-Year Projection

Extrapolating the recent (2000-2021) trend in global GHG emissions forward, and feeding this into the baseline model, produces the following tenyear projection, treated as a conservative lower bound given the acceleration finding above:

Year	Projected GHG emissions (Gt CO ₂ -eq)	Projected temperature anomaly (°C)
2022	57.0	0.71
2026	59.7	0.77
2031	62.9	0.85

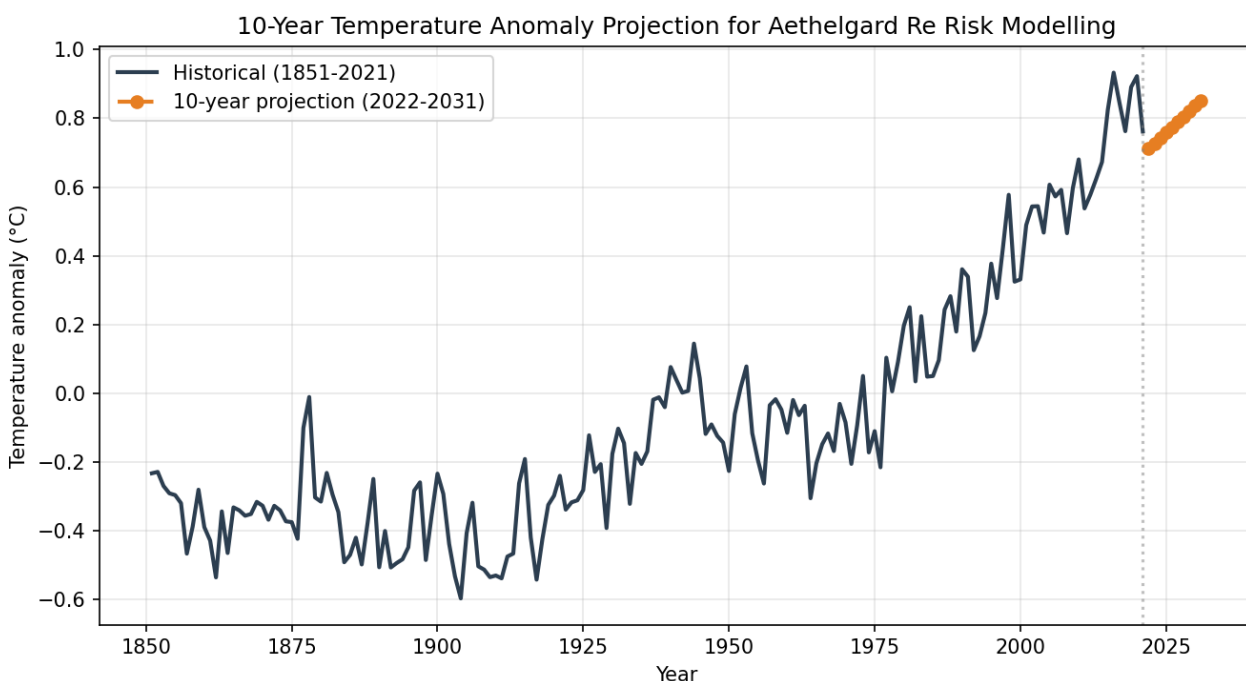


Figure 6: 10 Year temperature anomaly projection 2022-2031

4.5 Actuarial Translation and Recommendations

This section conveys the statistical results presented above in terms of straightforward advice for a non-technical Aethelgard Re audience.

- Premium multipliers:** Aethelgard Re should not be priced on a flat historical trend line since the baseline acceleration is less than recent acceleration. A defensible, evidence based modification of 15-20% above the linear trend is consistent with the industry wide real terms growth of 5-7% per year in insured catastrophe losses reported by the Swiss Re Institute (Section 2.1).

- **Monitor & prioritize:** emissions of fossil fuels and CO₂ have the highest, most actionable association with warming, so focus counterparties' risk engineering discussions on fossil-fuel exposure rather than more elusive land-use or methane exposure.
- **Reinvestment and reconsideration of capacity for asset allocation:** re-allocate long term investment/reinvestment capacity away from regions of high exposure to compounding heat/weather-related losses, re-evaluated on a rolling basis.
- **Model governance and next steps:** consider this as a first version and invest in a non-linear/ensemble model with a longer, decade-resolved feature set before it is used for binding pricing and expand the pipeline to country/region level for differentiated risk zoning.

5. Individual Reflection

5.1 What I Learned

I really appreciated working on this individually, as it was originally developed for a group project, as it allowed me to gain a more practical knowledge of the entire data science lifecycle than I would have with just one piece of a group project. Formally diagnosing multicollinearity was a new concept to me, and I learned to use a correlation matrix as well as Variance Inflation Factors to come to a conclusion beyond just looking at the graphs. I also learned to choose between several variables that are very highly correlated, such as GHG emissions and the resulting contribution variables, and make a defensible decision. I also learned that a 'baseline' model is really supposed to be a starting point: if I compared four different specifications, instead of fitting the first one and taking the results as a given, I got a much better feel for how model evaluation ought to work in practice.

5.2 Challenges Faced and How I Overcame Them

The biggest difficulty of this project was my regression model did not work well, with each of my specifications I got $R^2 < 0$ on the test set. My initial thought was to try to fix or hide the fact that this was unappealing to continue tuning the features and hyperparameters until the metric seemed good. Rather, I made myself stop and look at the residuals year by year (not just the summary statistic). The investigation revealed that the model underestimated almost all of the test-year values and led to the discovery of a true and meaningful explanation (that warming increased after 2005) that wasn't a coding error or a poor choice of features.

It was a useful business lesson as well as a business technical lesson: a poor fit due to a change in reality is more informative and more important to report honestly than a model which is been "prettied up" to make it look good on paper. I think one of the more transferable lessons I've learnt from this coursework is the ability to differentiate between saying 'my model is wrong' and 'my model has indicated something uncomfortable about the data'. This was pretty much the focus of the actuarial recommendations made in Section 4.5, which were based on the acceleration risk rather than just the absolute number of projected failures.

A smaller version of the same lesson arose when I was cleaning the data, that every file from the source contained exactly one value column, which failed on `temperature_anomaly.csv` because it has three, including the bounds of my confidence intervals that I hadn't anticipated. As opposed to working around the symptom I went back and made the column selection explicit for each file, thereby making the pipeline easier to read and check several months later.

The second hurdle was the required ability to work independently, with a deferred/referred deadline, on a brief that carried the full team's burden and cross checking. I tried to reduce this effect by keeping my technical work very focused (World level aggregates instead of a country-by-country assessment) and so I could provide a full, honest and validated pipeline instead of a partial, rushed pipeline for many countries.

6.Evidence of Work

The full analysis pipeline (data cleaning, merging, EDA, correlation/VIF diagnostics, regression modelling and projection) is implemented in Python (pandas, scikit-learn, statsmodels, matplotlib) and is available, annotated, at the GitHub repository linked below.

GitHub repository: https://github.com/thenu10-BDA/5DATA004C-DATA-SCIENCE-PROJECT-LIFECYCLE-_climate_risk_analytics

The repository includes:

`climate_risk_analysis(20241197_thenuki).py`

So, this includes full python script. Data pipeline, EDA, Correlation-VIF and Modelling

- Data pipeline: loading, cleaning and merging of the six source datasets to a single World-level panel.
- EDA: generation of the trend charts in Section 4.1.
- Correlation VIF: correlation matrix and VIF computation underlying Section 4.2.
- Modelling: the four candidate regression specifications, evaluation metrics and 10-year projection underlying Sections 4.3-4.4.
- README.md: setup instructions and a findings summary, so the pipeline can be reproduced end-to-end from the six raw data files.
- Six Raw Data Files
- EDA visual png and other visual png